

OptiCrop

Project Demo Screenshots

This document walks through OptiCrop's end-to-end flow — from entering soil and climate parameters to receiving an ML-generated crop recommendation — following the demo flow described in the project's **Demo Script** and the interface requirements in the **Software Requirement Specification**.

Note: No live screenshots of the running application were provided with the source documents. The screens below are illustrative mockups, built from the field names, sample values, validation rules, and model comparison figures documented in the uploaded project files (Demo Script, SRS, System Design, Final Report), to show how the application is intended to look and behave.

1	Field Data Entry Form	Input screen for N, P, K, temperature, humidity, pH, rainfall
2	Prediction Result	Recommended crop with confidence score
3	Input Validation	Error handling for missing / out-of-range values
4	Model Performance Summary	Accuracy comparison across 5 algorithms

1. Field Data Entry Form

The farmer enters the seven required parameters — Nitrogen, Phosphorous, Potassium, Temperature, Humidity, Rainfall and pH — through the web form built with HTML/CSS/Bootstrap (SRS §5.1). Sample values shown match the demo input set from the Demo Script (Step 3).



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STEP 1 - FIELD DATA ENTRY

Enter Soil & Climate Parameters

Provide the seven readings below and OptiCrop will recommend the best-fit crop for your field.

Field Parameters

All fields are required. Values should reflect current or recent soil test / weather station readings.

SOIL NUTRIENTS (MG/KG)

NITROGEN (N)

90

PHOSPHOROUS (P)

42

POTASSIUM (K)

43

CLIMATE CONDITIONS

TEMPERATURE (°C)

20.8

HUMIDITY (%)

82

RAINFALL (mm)

202

SOIL ACIDITY

PH LEVEL

6.5

► Predict Recommended Crop

Current Input Snapshot

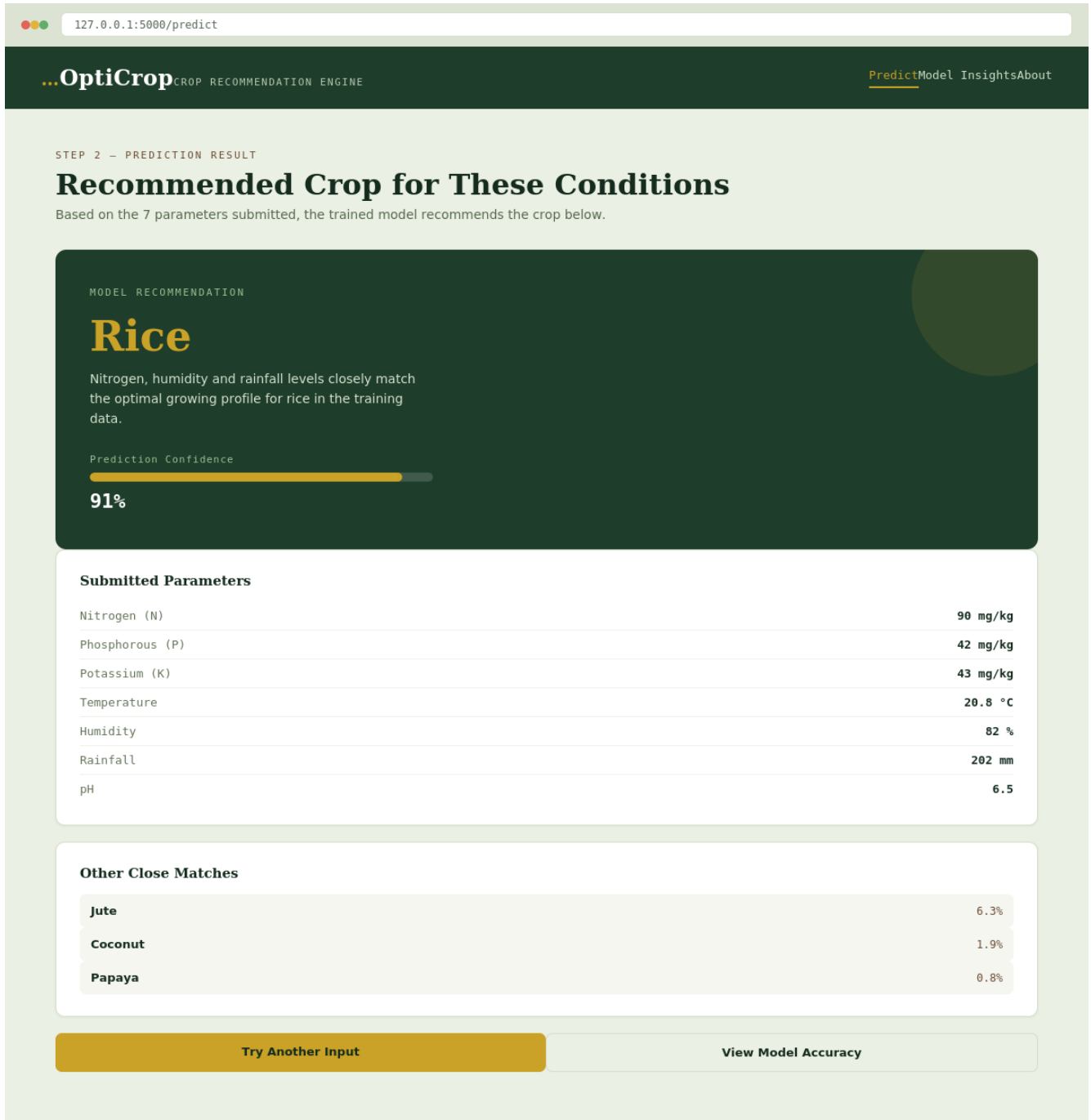
Nitrogen (N)	90 mg/kg
Phosphorous (P)	42 mg/kg
Potassium (K)	43 mg/kg
Temperature	20.8 °C
Humidity	82 %
Rainfall	202 mm
pH	6.5

HOW IT WORKS

Your inputs are validated, scaled, and passed to a pre-trained model (best of KNN, Logistic Regression, Decision Tree, Random Forest & K-Means) that returns the crop most suited to these conditions.

2. Prediction Result

After clicking Predict, the input is validated, preprocessed, and passed to the trained model. The result page displays the recommended crop (FR-5) along with a confidence score and the next-closest alternative crops.



3. Input Validation

Per FR-2, all fields are validated before processing. Blank or out-of-range values (e.g. humidity above 100%) are flagged inline with a clear message, and the prediction is blocked until corrected — matching the Edge Case Demo step in the Demo Script.



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INPUT VALIDATION

Some Fields Need Attention

FR-2: the system validates all fields before a prediction can be processed.

⚠ **Form not submitted — 2 issues found**
Fix the highlighted fields below, then predict again.

Field Parameters

All fields are required. Values should reflect current or recent soil test / weather station readings.

SOIL NUTRIENTS (MG/KG)

NITROGEN (N)

90

PHOSPHOROUS (P)

Required

⚠ This field cannot be blank

POTASSIUM (K)

43

CLIMATE CONDITIONS

TEMPERATURE (°C)

20.8

HUMIDITY (%)

140

⚠ Must be between 0 and 100

RAINFALL (MM)

202

SOIL ACIDITY

PH LEVEL

6.5

➤ Predict Recommended Crop

Validation Rules

Applied client- and server-side before any data reaches the model (SRS § FR-2).

N, P, K — numeric, required

Temperature — numeric, required

Humidity — 0–100%, required

Phosphorous — required, not blank

Rainfall — numeric, required

pH — 0–14, required

4. Model Performance Summary

Five algorithms — KNN, Logistic Regression, Decision Tree, Random Forest and K-Means — were trained and compared on the same dataset. Random Forest was selected for deployment based on its highest test accuracy, as summarized in the Demo Script (Step 5) and Final Report.

